Young-Sang Lee

Game server programmer, M.S. student

github.com/YSL-RyuDo | ysl-ryudo.github.io

Game server programmer building C++ servers and Unity multiplayer systems. I implement connection/session handling, game-state flows and client integration, with a focus on what must remain consistent across reconnects and asynchronous work. Currently pursuing an M.S. in Games (Engineering) at Hongik University; first author and presenter of two poster studies, and organizer/speaker of the inaugural I2P-CON.

Applied technical skills

C++ / Windows networking, IOCP, Winsock, sessions, packet dispatch, resource pools and DB worker separation (VAIS, Arcana Duel, Slap Squad).

C# / Unity, multiplayer client integration, match/profile/shop flows and data-driven gameplay systems.

Python / experiments, C2PA validation, perceptual-hash retrieval and simulated consumption analysis; test-source and recorded-result review.

Selected project experience

CodeName VAIS

Mar 2026 ~ Present, 10 members: 5 software, 5 graphics

Server development & Unity multiplayer integration | C++, Windows IOCP, C#, Unity

Implemented IOCP connection handling, session management and packet dispatch; connected room/match lifecycle, career/profile, shop and social services to Unity client contracts.

Separated reconnect presentation recovery from gameplay execution: ReconnectJournal keeps persistent state and excludes replay of damage, rewards and player commands, so restoring the view does not repeat those actions.

Implemented party/team voice authorization and reconnect-related client flows; integrated Sentinel abilities, lobby, results and player-facing UI with the networked game.

Case study | Server / Client, Private, access required

Arcana Duel

Aug 2025 ~ Feb 2026, 2 members: 1 programmer, 1 graphics

Programming: C++ server & Unity game integration | C++, IOCP, DB worker, Unity

Built the C++ IOCP server with receive/send workers and reusable session, buffer and overlapped-I/O resources; implemented the Unity game/network integration.

Assigned database requests to a separate DB worker, separating database execution from network completion handling.

Implemented magic/cooldown, match, profile and rank services, alongside damage/healing and ability integration for the multiplayer magic-combat game.

Case study | Server / Client, Private, access required

Slap Squad

Mar 2025 ~ Feb 2026, 8 members: 2 programmers, 6 graphics

Server, network integration & shop/profile systems | C++, Winsock, C#, Unity

Implemented the C++ Winsock server's account, room, shop and profile flows with connection threads and CSV-backed data; integrated the Unity client with the protocol.

Handled water-balloon and room-leader updates, purchase UI and the result/reward flow; linked authored client and server commits in the public repositories.

Developed a CNN study of fixed versus demand-adjusted pricing from the game's item-pricing question; presented it at I2P-CON, then refined the research for the CISC-W 2025 poster.

Case study | Client / Server, Public repository

Education & research experience

2026.09 ~ Present | M.S. student, Department of Games (Engineering)

Hongik University, Sejong Campus

2021 ~ 2026.08 | Bachelor's degree, Game Software

Hongik University, Sejong Campus

2025.03 ~ 2026.08 | Undergraduate researcher, Game Software

Hongik University, Sejong Campus

Research & presentations

CISC-W 2025, First author, poster presenter

2025 Winter Conference of the Korea Institute of Information Security and Cryptology

27 ~ 28 Nov 2025, Konjiam Resort, Gwangju, Korea

Consumption Pattern Analysis of In-Game Pricing Policies Using Convolutional Neural Networks

Compared pricing policies using 400 simulated players over 30 days, repeated 20 times. CNN consumption-pattern classification accuracy was 96%.

WISA 2026, First author, poster presenter

27th World Conference on Information Security Applications

26 ~ 28 Aug 2026, MAISON GLAD, Jeju, Korea

A Verification Method for Game Image Assets Using C2PA Manifest Validation and Perceptual Hash-Based Similarity Analysis

Evaluated C2PA validation and four perceptual hashes across 1,150 validation cases and 4,400 retrieval rows. Published code, results, paper and poster.

ICEA 2026, Accepted, presentation planned in Kaohsiung, research extensions in progress

International Conference on Intelligent Computing and its Emerging Applications

19 ~ 22 Nov 2026, Kaohsiung, Taiwan

Server-Side Multi-Parent Provenance Reconstruction in Heterogeneous Game MOD Packages

Research on multi-parent provenance retrieval using code, image and structured-resource features. Extending the accepted study while preserving its evaluation baseline.

Leadership & knowledge sharing

2025 inaugural I2P-CON, Operations Team Lead & Speaker

Recruited the organizing team and led operations for the inaugural student-led game-development conference, jointly hosted by I2P and O2CUBE.

Presented the CNN pricing-policy research in “Designing Game Economies with Data.” Subsequently refined the research for the CISC-W 2025 poster presentation.

Produced a conference publication and distributed it to current students and alumni. The official playlist preserves eight talks.

My talk | I2P-CON

O2CUBE, Software Team Lead, 2025

Led C programming and computer architecture study sessions from March to June 2025. Covered C fundamentals through arrays, and guided a semester of computer architecture study.

Additional engineering

DeathGuider: management, facilities and battle-preparation systems. PitchLab: bounded FIFOs and DSP worker separation.

I participate in software development at Hongik University's Game Content Copyright Technology Global Lab (GCT Lab).
GCT Lab